CEDAR Madrigal Database

TEC Maps Query

User Manual — Global Configuration

Instrument 8000 | GPS/GNSS Receivers
Full Earth | 5-minute cadence
8 sample maps at 3-hour intervals
cedar.openmadrigal.org

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1. Overview

This manual describes how to use tec_maps_global.py to retrieve and visualize Total Electron Content (TEC) maps from the CEDAR Madrigal database for the entire Earth. No spatial filter is applied — every valid 1°×1° grid cell globally is returned and plotted. The TEC map preserves the full 2-D spatial distribution of ionospheric electron content at a single moment in time.

1.1 What is a TEC Map?

A TEC map is a 2-D snapshot of the ionosphere: a grid of TEC values (in TECU) at every 1°×1° latitude–longitude cell at one specific 5-minute timestamp. Plotted as a color image, it shows how TEC varies across a geographic region — high over the sunlit hemisphere, low over the dark side, with gradients controlled by the solar zenith angle, geomagnetic latitude, and space weather conditions. A TEC map is a snapshot at one moment in time of the ionosphere's state across a given region. The map is a picture detailing where the ionosphere is "thick" (dense with electrons, high TEC) and where it is "thin" (low TEC) at a particular instant.

1.2 Relationship Between TEC Maps and TEC Time Series

Note: ***Maps and time series both come from the exact same Madrigal file and the same isprint API call. The only difference is how you group the returned rows.***

The Madrigal binned file (instrument 8000, gpsYYMMDDg.hdf5) stores TEC as a 3-D data cube:

- Dimension 1: Time — 288 5-minute timesteps per day (00:02:30, 00:07:30, …, 23:57:30 UTC)
- Dimension 2: Latitude — 180 cells, 1° each, from −90° to +90° N
- Dimension 3: Longitude — 360 cells, 1° each, from −180° to +180° E

This gives roughly 180 × 360 × 288 = ~18.7 million cells per file, though many are flagged as missing (no GPS coverage). When you query isprint with a spatial filter and no time filter, you receive all three dimensions in the filtered region — one row per (time, lat, lon) combination.

The two products (time series vs maps) differ ONLY in how that table of rows is sliced:

Product	What is fixed	What varies	How many rows per fetch
Time Series	Geographic cell (lat, lon)	Time (5-min steps)	~288 per day (one median per timestep)
TEC Map	Timestamp	Lat, Lon (all cells in region)	~18,000–45,000 per map (full global grid, varying by GPS coverage)

In tec_maps_global.py, all 288 daily timesteps for the full Earth are fetched in a single isprint call (no spatial filter). The script then slices that dataset 8 times — once per chosen timestamp — to extract the global map for each sample time and render it as a 2-D color plot.

1.3 Physical Meaning of Each Data Value

Each time point in the time series are the TEC values for the four 1°×1° grid cells whose center is at 40.5°N, −73.5°E at each 5-minute timestamp. The cells contain NYC, but it's not an average of multiple cells. Rather, it's four different individual cell's values. If you go to tec_global_data.csv, you can see that at every 5 minute interval, the 4 cells that make up NYC record their individual TEC value. Each value is the number of electrons per square meter in a vertical column extending from the ground through the entire ionosphere above that 1°×1° patch of ground, averaged over all GPS measurements passing through that cell during that 5-minute window

2. Sample Map Selection

The script selects 8 map snapshots spaced evenly at 3-hour UTC intervals: 00:00, 03:00, 06:00, 09:00, 12:00, 15:00, 18:00, and 21:00 UTC. These 8 times divide the 24-hour day into equal segments and capture every major phase of the ionospheric diurnal cycle.

2.1 The 8 Selected Maps

Panel	UTC Hour	UTC−5 (EST) approx.	Ionospheric Phase	Why Selected
1	`00:00`	19:00 (prev)	**Post-sunset decay**	Solar EUV has been off for ~3 h. TEC has been falling since ~22 UTC. Captures the early nighttime low.
2	`03:00`	22:00	**Deep nighttime minimum**	TEC typically reaches its daily minimum near local midnight. Lowest values of the day expected.
3	`06:00`	01:00	**Pre-dawn trough**	Recombination continues through the night. This is often the true TEC minimum, just before sunrise ionization begins.
4	`09:00`	04:00	**Sunrise ionization onset**	Solar terminator crosses 74° W (~10:30 UTC). By 09 UTC the ionosphere is just beginning to be ionized. Rapid TEC rise imminent.
5	`12:00`	07:00	**Morning build-up**	~2 hours after sunrise. TEC rising steeply. Clear asymmetry between day-side and night-side visible in map.
6	`15:00`	10:00	**Late morning peak approach**	TEC typically peaks in the late morning to early afternoon at mid-latitudes. Often the highest values of the day.
7	`18:00`	13:00	**Post-noon plateau**	TEC plateau or gradual decline after peak. Day–night gradient visible but less steep than at 12 UTC.
8	`21:00`	16:00	**Afternoon / pre-sunset**	Sun still above horizon. TEC still elevated. Within ~2 h of sunset onset at this longitude.

The 8 maps together form a complete picture of one diurnal cycle. Reading the panels left-to-right, top-to-bottom corresponds to stepping forward in time and watching the ionosphere respond to the rising and setting sun. On a global map every panel shows the full day–night terminator simultaneously. The 06:00 and 09:00 UTC panels are particularly striking because the terminator crosses the Americas, making the hemispheric TEC contrast especially sharp.

2.2 How the Nearest Timestamp is Selected

Because the binned data uses 5-minute mid-point timestamps (00:02:30, 00:07:30, …), there is no record exactly on the hour. The script calls nearest_timestamp(), which finds the unique datetime in the dataset with the smallest absolute time difference from the target (e.g., 00:00:00 UTC matches 00:02:30 UTC, a 2.5-minute offset). This offset is printed to the console at runtime for transparency.

3. Setup and Installation

Setup is identical to the TEC time series scripts. If you have already created the tec-env virtual environment, activate it and skip to step 4.

1. Create the virtual environment (once only):
```
python3 -m venv ~/tec-env
```

2. Activate it (every new terminal session):
```
source ~/tec-env/bin/activate
```

3. Install packages (once only, after activation):
```
pip install madrigalWeb pandas matplotlib numpy
```

4. Run the script:
```
python3 tec_maps_global.py
```

5. View outputs:
```
xdg-open tec_maps_global_8panel.png
```

4. Configuration

4.1 Required: User Credentials

```
USER_NAME        = "Peter Daudelin"
USER_EMAIL       = "pd475@njit.edu"
USER_AFFILIATION = "NJIT"
```

4.2 Query Parameters

Parameter	Default Value	Description
MADRIGAL_URL	`https://cedar.openmadrigal.org`	Server URL. Do not change for CEDAR.
INST_CODE	`8000`	Instrument code for GPS/GNSS Receivers.
START_DATE	`datetime(2023, 1, 1)`	Start of query window. For maps, one day is sufficient.
END_DATE	`datetime(2023, 1, 1)`	End of query window. Same as START_DATE for a single-day map set.
FILE_FILTER	`"binned"`	Selects 1°×1°×5-min binned HDF5 files.
PARAMETERS	`"gdlat,glon,tec"`	Parameters fetched. gdlat/glon are grid centers; tec is vertical TEC in TECU.
SAMPLE_HOURS	`[0,3,6,9,12,15,18,21]`	UTC hours of the 8 selected map times. Change to shift sampling.
OUTPUT_CSV	`"tec_maps_global_data.csv"`	Full regional dataset for all timesteps.
OUTPUT_8PANEL	`"tec_maps_global_8panel.png"`	8-panel figure PNG.

4.3 Global Map Extent — No Spatial Filter

No spatial filter is applied. The full Earth — all latitudes from −90° to +90° N and all longitudes from −180° to +180° E — is fetched and plotted. The FILTER_STR constant is set to an empty string in the script. This makes the fetch significantly larger than a regionally filtered query (approximately 4–5 million rows per file), so expect a runtime of 5–15 minutes per file.

Latitude coverage	−90° to +90° — full global latitude range

Longitude coverage	−180° to +180° — full global longitude range

The color scale is fixed at 0–60 TECU across all 8 panels. The upper limit is raised to 60 TECU to accommodate the equatorial ionospheric anomaly, which regularly exceeds 40 TECU in daytime. White reference lines mark the equator and prime meridian on every panel.

5. Output Files

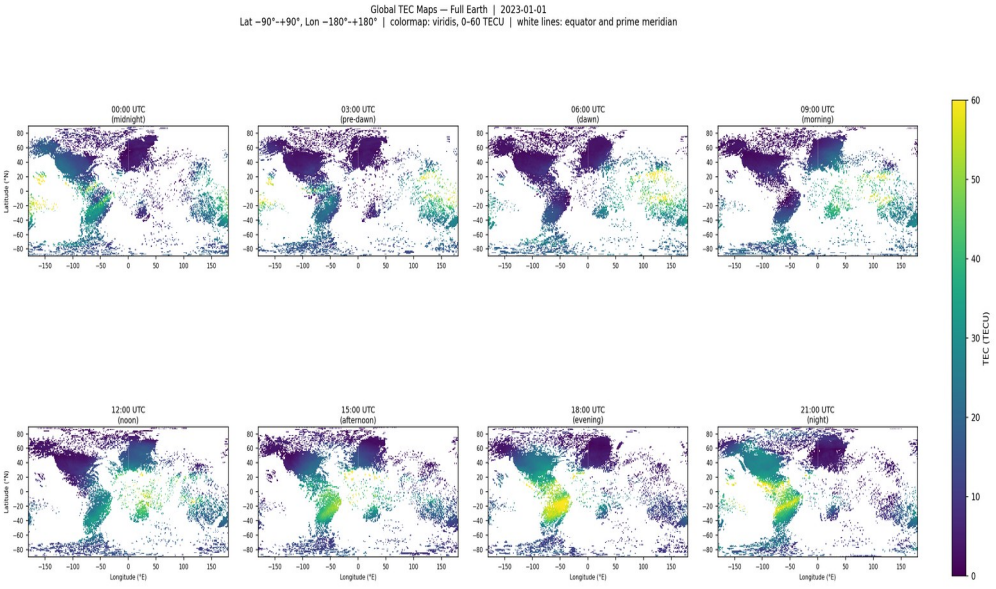
5.1 tec_maps_global_data.csv

Contains every row returned by the isprint query — all 288 timestamps × all globally valid grid cells. Approximately 4–5 million rows for a single day **(not 18 million, because blank cells do not get recorded)**. Columns: year, month, day, hour, min, sec, gdlat, glon, tec, datetime. This file contains all the data needed to extract any map time, compute zonal statistics, or build animations.

	A	B	C	D	E	F	G	H	I	J
1	year	month	day	hour	min	sec	gdlat	glon	tec	datetime
2	2022	12	31	0	2	30	-90	-87	13.272	2022-12-31 00:02:30
3	2022	12	31	0	2	30	-90	-64	12.3095	2022-12-31 00:02:30
4	2022	12	31	0	2	30	-90	-63	12.3832	2022-12-31 00:02:30
5	2022	12	31	0	2	30	-90	-62	12.2604	2022-12-31 00:02:30
6	2022	12	31	0	2	30	-90	-60	12.2574	2022-12-31 00:02:30
7	2022	12	31	0	2	30	-90	-59	12.2541	2022-12-31 00:02:30
8	2022	12	31	0	2	30	-90	-58	12.3012	2022-12-31 00:02:30
9	2022	12	31	0	2	30	-90	-56	12.3154	2022-12-31 00:02:30
10	2022	12	31	0	2	30	-90	-55	12.2614	2022-12-31 00:02:30
11	2022	12	31	0	2	30	-90	-53	12.2854	2022-12-31 00:02:30
12	2022	12	31	0	2	30	-90	-52	12.2386	2022-12-31 00:02:30
13	2022	12	31	0	2	30	-90	-50	12.2754	2022-12-31 00:02:30
14	2022	12	31	0	2	30	-89	-131	12.5788	2022-12-31 00:02:30
15	2022	12	31	0	2	30	-89	-130	12.5177	2022-12-31 00:02:30
16	2022	12	31	0	2	30	-89	-129	12.572	2022-12-31 00:02:30
17	2022	12	31	0	2	30	-89	-128	12.5713	2022-12-31 00:02:30
18	2022	12	31	0	2	30	-89	-93	11.1617	2022-12-31 00:02:30
19	2022	12	31	0	2	30	-89	-91	11.1825	2022-12-31 00:02:30
20	2022	12	31	0	2	30	-89	-89	11.109	2022-12-31 00:02:30
21	2022	12	31	0	2	30	-89	-87	13.0662	2022-12-31 00:02:30
22	2022	12	31	0	2	30	-89	-41	13.3458	2022-12-31 00:02:30
23	2022	12	31	0	2	30	-89	-38	13.3914	2022-12-31 00:02:30
24	2022	12	31	0	2	30	-89	-37	13.4055	2022-12-31 00:02:30
25	2022	12	31	0	2	30	-89	-34	13.4274	2022-12-31 00:02:30

5.2 tec_maps_global_8panel.png

A single 22×10 inch figure with 8 panels arranged in 2 rows × 4 columns, one per sample time. All panels show the full globe (−90° to +90° lat, −180° to +180° lon). All panels share the same viridis colormap scaled 0–60 TECU, with a single shared colorbar on the right edge. White lines mark the equator and prime meridian on every panel.



5.3 tec_map_global_HHMM.png (8 individual files)

One 10×5 inch PNG per sample time, named by UTC hour (e.g. tec_map_global_0000.png, tec_map_global_0300.png, …, tec_map_global_2100.png). The wider aspect ratio accommodates the full −180° to +180° longitude span. Each has its own colorbar and axis labels.

Example consol output:

```
(tec-env) peter@daud:~/uri/files/global/maps$ python3 tec_maps_global.py
Connecting to https://cedar.openmadrigal.org ...
Searching instrument 8000 2023-01-01 (CEDAR only) ...
Found 2 experiment(s).

Querying 2 file(s) [binned] — NO spatial filter (global):
  gps221231g.003.hdf5  [TEC binned 1 degree by 1 degree by 5 min]
  gps230101g.002.hdf5  [TEC binned 1 degree by 1 degree by 5 min]

Fetching gps221231g.003.hdf5 (global — this may take several minutes) ...
  -> 4711463 records

Fetching gps230101g.002.hdf5 (global — this may take several minutes) ...
  -> 4374600 records

Total records     : 9086063
Time range        : 2022-12-31 00:02:30  ->  2023-01-01 23:57:30
Unique lat cells  : 180
Unique lon cells  : 360
TEC (TECU)        : min=0.00  median=13.00  max=296.05

Saved -> tec_maps_global_data.csv

Selecting 8 sample map times (3-hour intervals) ...
  00:00 UTC target -> nearest record: 00:02:30
  03:00 UTC target -> nearest record: 03:02:30
  06:00 UTC target -> nearest record: 06:02:30
  09:00 UTC target -> nearest record: 09:02:30
  12:00 UTC target -> nearest record: 12:02:30
  15:00 UTC target -> nearest record: 15:02:30
  18:00 UTC target -> nearest record: 18:02:30
  21:00 UTC target -> nearest record: 21:02:30

Plotting 8-panel global TEC map figure ...
  Saved -> tec_map_global_0002.png
  Saved -> tec_map_global_0302.png
  Saved -> tec_map_global_0602.png
  Saved -> tec_map_global_0902.png
  Saved -> tec_map_global_1202.png
  Saved -> tec_map_global_1502.png
  Saved -> tec_map_global_1802.png
  Saved -> tec_map_global_2102.png

Saved -> tec_maps_global_8panel.png
```

6. Data Cadence and Grid Details

Cadence	5 minutes — 288 timestamps per day (00:02:30 to 23:57:30 UTC)
Spatial grid	1°×1° cells; cell label = center coordinate
Cell coverage	All cells: −90.5° to +90.5° N, −180.5° to +180.5° E
Global cells per timestamp	Up to 180 × 360 = 64,800 cells (fewer due to missing GPS coverage)
Rows per full-day global fetch	~4–5 million rows per file (varies by GPS coverage)
Missing value sentinel	1×10^{30} — replaced with NaN by script
Color scale	0–60 TECU, fixed across all 8 panels; raised ceiling for equatorial anomaly

7. Troubleshooting

Blank or all-NaN map panels	Some global cells will always be NaN over ocean areas with sparse GPS receivers. This is expected. If entire panels are blank, the file may be corrupt — try a different date.
TimeoutError	Ensure local=1 is set in getExperiments. The default local=0 queries all federated sites and times out.
pcolormesh shape error	Occurs if a map timestamp has incomplete lat/lon coverage. The pivot() call in plot_map() will have NaN-filled holes — these render as blank cells, which is expected.
Only 1–2 sample maps appear	Ensure START_DATE = END_DATE for a single-day fetch. Multi-day global queries can produce very large CSVs and long runtimes.
UserWarning: FigureCanvasAgg non-interactive	Harmless. PNG files are still saved. Use xdg-open to view them.

8. Extending the Script

8.1 Add More Sample Times

Edit SAMPLE_HOURS to include more entries (e.g. every hour: list(range(24))). Update SAMPLE_LABELS to match. **The 8-panel layout is hardcoded as 2×4.** For a different number of panels, change the gridspec call.

8.2 Animate All 288 Timesteps

To create an animation of the full day, loop over all unique datetime values in the CSV, call plot_map() for each, save a PNG, and combine with ffmpeg:

```
ffmpeg -r 10 -pattern_type glob -i 'tec_map_*.png' -vcodec libx264 tec_day.mp4
```

8.3 Change the Color Scale

The vmin=0, vmax=60 range covers quiet to moderately active conditions globally. During major geomagnetic storms the equatorial anomaly can exceed 100 TECU; increase vmax accordingly.

8.4 Compare to the Time Series

The global median computed from tec_maps_global_data.csv at each timestamp should match the global median time series produced by query_tec_global.py. You can verify this by grouping the maps CSV by datetime, computing the median tec, and plotting it — it should reproduce the time series plot exactly.

9. Citation

Per the CEDAR Rules-of-the-Road, use of this data in publications requires:

6. Contact the data PI (Anthea Coster, acoster@haystack.mit.edu) before formal use.
7. Acknowledge the CEDAR Madrigal database in all reports and publications.

```
Rideout W., Cariglia K. CEDAR Madrigal Database URL:
https://cedar.openmadrigal.org
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